# AI Image Generator

Appendix

Version 1.0
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Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

# Appendix

## A. Project file tree

```
ai-image-gen/
  app.py                 web application (Gradio)
  prompt_parser.py       prompt -> settings
  upscaler.py            Real-ESRGAN tiled upscaling
  gpu_info.py            GPU facts and live telemetry
  db.py                  SQLite store and CLI viewer
  generate.py            command-line generator
  requirements.txt
  run-app.bat
  README.md
  generations.db         created on first generation
  models/
    RealESRGAN_x4plus.pth
  outputs/               generated PNG files
  docs/
    00-...-Complete-Technical-Documentation.pdf
    01-Project-Overview.pdf ... 11-Appendix.pdf   one PDF per section
  docs-src/
    01-overview.md ... 11-appendix.md   editable sources
    images/
    build_pdf.py
  .venv/                 virtual environment (not for backup)
```

## B. requirements.txt

```
# Install PyTorch with CUDA first (RTX 3090 / any recent NVIDIA card):
#   pip install torch --index-url https://download.pytorch.org/whl/cu126
# Then:
#   pip install -r requirements.txt
diffusers>=0.31
transformers>=4.45
accelerate>=1.0
safetensors
gradio>=5.0
spandrel>=0.4
# torchvision: pip install torchvision --index-url https://download.pytorch.org/whl/cu126
```

## C. Command cheat sheet

All commands run from the project folder in PowerShell.

| Task | Command |
|---|---|
| Start the app | `.\.venv\Scripts\python.exe app.py` |
| Generate from CLI | `.\.venv\Scripts\python.exe generate.py "prompt" --seed 42 --count 2` |
| Upscale a file | `.\.venv\Scripts\python.exe upscaler.py outputs\x.png 3840` |
| List history | `.\.venv\Scripts\python.exe db.py` |
| One record | `.\.venv\Scripts\python.exe db.py 4` |
| Statistics | `.\.venv\Scripts\python.exe db.py stats` |
| Parser self-test | `.\.venv\Scripts\python.exe prompt_parser.py` |
| GPU telemetry dump | `.\.venv\Scripts\python.exe gpu_info.py` |
| Rebuild the PDFs | `.\.venv\Scripts\python.exe docs-src\build_pdf.py` |
| Check CUDA | `.\.venv\Scripts\python.exe -c "import torch; print(torch.cuda.is_available())"` |

## D. Programmatic access to the running app

```
from gradio_client import Client
c = Client("http://127.0.0.1:7860")
images, report = c.predict("a lighthouse, 16:9, 4k, seed 42", "", api_name="/generate")
table, stats = c.predict("lighthouse", 50, api_name="/refresh_history")
```

## E. Licences

| Component | Licence |
|---|---|
| Stable Diffusion XL base 1.0 | CreativeML Open RAIL++-M |
| Real-ESRGAN weights and code | BSD 3-Clause |
| diffusers, transformers, accelerate | Apache 2.0 |
| Gradio | Apache 2.0 |
| spandrel | MIT |
| PyTorch | BSD-style |
| Project source code | Owned by Frazile AI |